

Shivang Dubey

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EDUCATION

- **Indraprastha Institute of Information Technology Delhi** IIIT Delhi
Post Graduate Diploma in Data Science in Health and Climate Change for Social Impact Oct 2025 - Jul 2026
- **Guru Gobind Singh Indraprastha University** BPIT Delhi
B.Tech. in Electronics and Communication Engineering; CGPA: 9.48/10.00 Aug 2019 - Jul 2023

RESEARCH EXPERIENCE

- **Indraprastha Institute of Information Technology Delhi** IIIT, Delhi, IN
Research Assistant, mentor Prof. Saket Anand Jul 2023 - Aug 2023
 - **Highway Traffic Analysis and Object Tracking:** Conducted research at the **Networked Autonomous Systems Lab** on highway traffic analysis, developing and optimizing computer vision algorithms for object detection, tracking, and segmentation across traffic video data.
 - **Video Data Preprocessing:** Built preprocessing workflows for **.rec** video files, including data extraction, transformation, and preparation to ensure reliable input for downstream computer vision and traffic analysis pipelines.
 - **Image Data Decoding:** Developed methods to decode and process raw video data for integration into the lab's analysis pipeline, enabling frame-level analysis, object tracking, and extraction of information for studying **traffic patterns and vehicle movement**.
- **Polytechnique Montréal** Montréal, Canada
Research Intern, mentor Prof. Soumaya Cherkaoui May 2022 - Jul 2022
 - **Reinforcement Learning:** Collaborated with a PhD researcher to develop a Deep Q-Learning (DQN) model for a Network Intrusion Detection System (NIDS), achieving **79%** classification accuracy on the target dataset.
 - **Comparative Analysis:** Developed and evaluated multiple machine learning approaches for the same NIDS, including Support Vector Machine (SVM), Random Forest, and Convolutional Neural Network (CNN) models, and benchmarked their performance against the DQN model using **TensorFlow**.
 - **Outcome:** Conducted a comparative performance analysis and found that the Deep Q-Learning model outperformed the other evaluated ML approaches by approximately **2%** in accuracy on the same dataset.
- **Indian Institute of Technology Delhi** Remote
Research Intern, mentor Prof. Biswajita Parida Jun 2021 - Aug 2021
 - **Qualitative Data Analysis:** Researched over **150+ companies/start-ups** to identify the reasons behind their products' successes and failures in the market. Modelled a case study on a product to help boost its reach through **numerical and qualitative analysis**.
 - **Case Study:** Mined and examined data related to **consumer psychology** and **neuromarketing** while working on the project "Consumer adoption behaviour of Ayurvedic offerings in the market."
 - **Implementation:** Forecasted an increase in sales by applying the '**Segmentation, Targeting and Positioning**' model and the **4P** (Product, Price, Place, Promotion) framework.

INDUSTRY EXPERIENCE

- **ZS Associates** Gurugram, India
Business Technology Solutions Associate Consultant Sep 2023 - Present
 - Developed and executed scalable ETL processes that converted **500+** raw datasets into actionable sales and compensation reports, implementing rigorous data validation measures to achieve **99%** data accuracy across systems.
 - Conducted routine data manipulation and insight generation using SQL (PostgreSQL, Workbench). Managed data ingestion and extraction via Amazon S3, and processed large healthcare datasets including medical sales and claims data exceeding **5M+** records.
 - Developed performance analytics dashboards in Power BI and MicroStrategy for a field force of **1,000+** healthcare representatives, enabling individual performance tracking and contributing to a **25%** increase in goal attainment.
 - Automated backup processes on Amazon S3 using Python and **boto**, reducing manual effort by approximately **8 hours** per week. Streamlined operations and reporting workflows using Apache Airflow, saving an additional 4 hours weekly.
 - Developed Excel automation solutions using VBA, eliminating a **12-hour** weekly manual process for data fetching, formatting, and analysis. Created VBA-driven Excel reports linked to PostgreSQL databases, enhancing team efficiency by approximately **80%**.
 - Led backend engineering for a web portal as part of a critical technology migration project, implementing complex business logic with Python and Flask and integrating with an AngularJS frontend. The platform supports **1,000+** users in a US healthcare marketing ecosystem managing data for **4** key pharmaceutical products.

- Managed and mentored a team of **4 Associates**, overseeing onboarding, conducting regular knowledge transfer sessions, and providing technical guidance on ETL development, SQL, and data workflows — ensuring consistent delivery standards across the team.
- Collaborated directly with client stakeholders to gather requirements, translate business needs into technical specifications, and deliver reporting solutions aligned to sales and compensation workflows.

PROJECTS

- **Social Distancing Violation Detection System (Academic)** Dec 2022
Python, Computer Vision, Human Detection [Source](#)
 - **Working:** The model was created to detect humans and distance violations between two human centroids, along with human face detection.
 - **Utilization:** Model trained using the **YOLO** (v3) algorithm and weights, along with the COCO dataset for pre-existing object classes. **OpenCV** for face and human detection, along with **SciPy**'s Euclidean distance for human centroid formation.
 - **Outcome:** With a suitable detection rate, the model draws a box around the human; accuracy more than **94%**.
- **COVID-19 Detector** Jun 2021
Python, Machine Learning, Computer Vision [Source](#)
 - **Working:** COVID-19/infected cases were detected using a positive case and a normal case image dataset. **Gradient Descent** and a CNN classifier were used to optimize the process.
 - **Utilization:** Deep learning and model training using **Keras** and the **CNN** algorithm. Accuracy more than **90%**.
- **Online Grocery Store Scraper** Jul 2020
Python, Data Scraping, Mining, Automation [Source](#)
 - **Working:** Data was scraped from all products on the online grocery store, including price, description, quantity, and brand, and saved in a reusable format for data analysis and operations.
 - **Utilization:** BeautifulSoup4 (**bs4**) was used to scrape the data, and **Selenium** was used to automate the browser.

TECHNICAL SKILLS

- **Languages:** Python, SQL (MySQL, PostgreSQL), C/C++
- **Tools & Frameworks:** pgAdmin, NumPy, Pandas, OpenCV, Flask, Git, GitHub
- **BI & Reporting Tools:** Power BI, MicroStrategy, Excel, PowerPoint
- **ETL & Orchestration:** Apache Airflow, IICS
- **Cloud & Data Platforms:** AWS (S3, EC2, ECS, Redshift), Snowflake
- **AI Tools:** Claude, ChatGPT, Gemini (prompt engineering for development & analytics workflows)
- **Key Skills:** Data Analysis, Data Visualization, ETL Development, Data Ingestion, Backend Development, Machine Learning, Healthcare Data, Computer Vision, Segmentation and Classification

AWARDS AND ACHIEVEMENTS

- **Promotion to Associate Consultant:** Promoted from Associate for consistent high quality delivery
- **ZS Impact Award:** Recognized for high-quality, expertise driven delivery and a commitment to excellence
- **Academic Distinction:** Graduated **2nd** of 120, with top-rank standing in the second and third years
- **Mitacs Globalink Research Internship:** Fully funded research internship position at a Canadian university
- **Smart India Hackathon 2022 College Finalist:** Top-10 in college to represent the team in additional SIH rounds
- **Break the Shackles 3.0 Finalist:** Finalist in inter-college technical hackathon organised by the college and IEEE

CO-CURRICULAR ACTIVITIES

- **Mentor:** #define; College Technical Society Aug 2020 - Sep 2022
- **Volunteer:** PyCon India 2021; annual global conference for Pythonistas Feb 2021 - Sep 2021
- **Author:** [SymPy](#); Python open-source library for symbolic mathematics
- **Talk:** Getting Started with Open Source, in collaboration with GirlScript Foundation Jan 2021
- **Talk:** Introduction to Open Source with Git and GitHub Sep 2020
- **Academic Counsellor:** [Careers360](#) May 2020 - Aug 2020

COURSES UNDERTAKEN

- **Core:** Analog Electronics, Digital Communication, Digital Signal Processing, Computer Architecture and Organisation, Database Management System, Data Structures and Algorithms, Signals and Systems, Logic Design, Network Synthesis, Control System, Microprocessors and Micro Controllers, Very Large Scale Integration Design
- **Non-Core:** Python, [Structured Query Language \(SQL\)](#), [Natural Language Processing \(NLP\)](#), [Applied Text Mining](#), Machine Learning, [Data Science with Python](#), Data Analysis, [Data Structures with C++](#)